PROBLEM STATEMENT 03 · QDRANT EDGE

FieldMind

Memory for field technicians that works with no network, knows what it is allowed to share, and syncs with the fleet whenever a link exists.



Team Compilers · github.com/yabhinav1/fieldmind

The most useful field knowledge is created where the network is worst

1

No signal

Plant floors, basements and remote sites drop the connection for hours. The work does not stop.

2

Mixed notes

One note can hold a fault the whole fleet needs and a colleague's name that must never leave the device.

3

Drift

Devices that work apart come back with different versions of the truth, and someone's edit gets lost.

FieldMind is one process on the device: Qdrant Edge for memory, small on-device models for meaning, names, reranking, speech and photos, SQLite for what is waiting, and a dashboard a technician can use with gloves on. Two devices run 24/7 on arm64 servers against Qdrant Cloud; the same code runs on a laptop or a Raspberry Pi.

The device that keeps working

1

Record

Write, dictate or photograph what you saw. Hands-free dictation in any language, transcribed on the device by Whisper; the device shows what it will share before you save, and learns from your overrides.

2

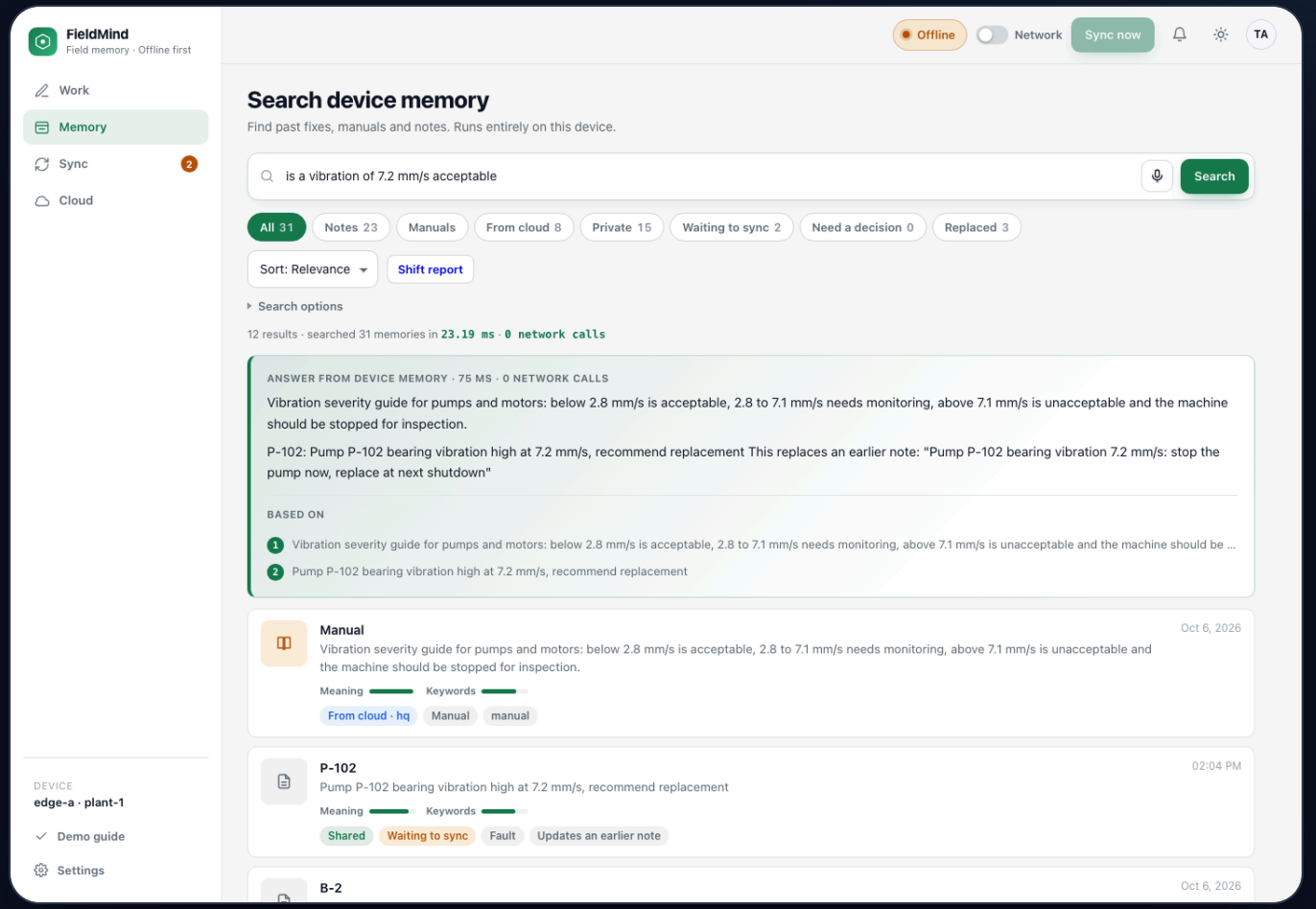
Find

Hybrid search over your notes, your photos and headquarters manuals in about a millisecond. A cross-encoder rereads the best candidates.

3

Ask

A cited answer written on the device. Every number, asset and verdict is checked against the note it cites. No cloud call.



One device process, one cloud collection

EDGE DEVICE

Local shard
Written here, private too. Sealed on disk with the device key.

Replica shard
Cloud knowledge, or what nearby devices shared when the cloud was gone.

Policy engine
Decides what may leave. Remembers the person's overrides as vectors, never text.

Embeddings
bge-small (or multilingual) and BM25; MiniLM reranker; CLIP for photos.

Journal
Outbox, conflicts, sealed activity. Bounded, survives restarts.

Answer model
llama3.2 3B, optional, with a source check on every answer.

Push →
compare and swap, payload-only when the text is unchanged

Pull ←
diff, or one snapshot when far behind

QDRANT SERVER

Shared notes
Masked notes
Headquarters manuals
Photo thumbnails

Other devices sync here too, and with each other over the local network when it is unreachable.

The device decides what leaves

WHAT THE TECHNICIAN WROTE	WHAT HAPPENS
Pump P-102 bearing vibration high at 7.2 mm/s	Shared
Smell of gas near compressor K-4	Shared · Urgent, sent first
Suresh and Priya replaced the coupling, call 9876543210	Shared, names and number masked
Technician Ravi Kumar reported chest pain	Private, stays on the device
SCADA panel login password is …	Private, cannot be overridden
Operator slipped on oil near P-102, no injury	Shared · Flagged: reads as two kinds

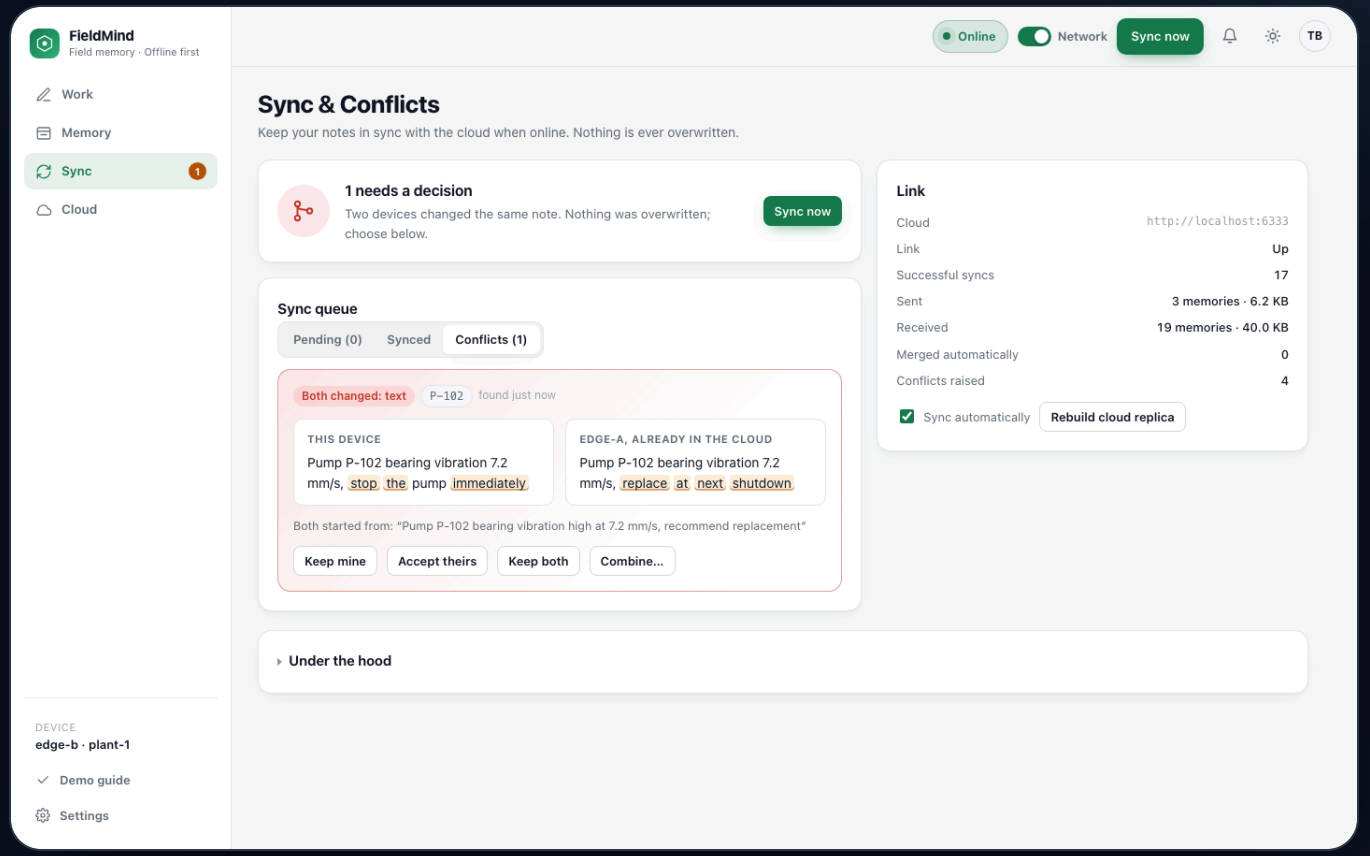
About 20 ms on the device: a name recognition model, pattern detectors for credentials, contacts, identity numbers, vehicles, badges and addresses, and a semantic classifier. Vectors for a masked note are computed from the masked text. Override a decision and notes like it follow your choice from then on. A trainee device's shared notes wait for a supervisor's approval; safety hazards never wait.

When two devices disagree

1. **Nothing is overwritten**
   An upload lands only if the cloud is still at the revision the device last saw, confirmed by a per-write id. An edit made mid-upload is rebased, not lost.

2. **Most edits merge**
   Changes to different fields combine automatically, field by field.

3. **A person settles the rest**
   Keep mine, accept theirs, keep both or combine. Even while offline. One click stages a conflict for the demo.



What Qdrant does for us

QDRANT FEATURE	HOW FIELDMIND USES IT
Edge shards, in process	Three per device: local notes, a cloud replica, and photos with a CLIP vector each
Built-in BM25 with IDF	Keyword half of hybrid search, with no server
Dense vectors and filters	Meaning search narrowed by asset, scope and status; photos found by what they show
Shard snapshots	A device far behind restores its replica from one file
Conditional upsert and payload overwrite	Compare and swap, so devices never clobber each other; payload-only patches skip the vectors
One schema on both sides	The same dense and sparse vectors on edge and server, versioned payloads

qdrant-edge-py 0.8.0 with Qdrant Server 1.19.2, pinned and tested in CI.

Measured, not estimated

1 ms
hybrid search across both shards

0
network calls to search, answer, or find a photo

20 ms
to decide what may leave the device

1–3 s
for a cited answer from a 3B model

132
automated tests on every push: two devices, a real Qdrant Server, a real browser

30
checks in the live end-to-end run

Ryzen 7 7435HS laptop, RTX 3050 4 GB. Also verified on Linux through a real network disconnect, and in production on two arm64 servers with Qdrant Cloud. Dictation: about 0.4 s for a six-second clip on a CPU.

Five moments in the demo

1. Switch the network off and keep working
2. Ask a question, get a cited answer
3. Watch names get masked before upload
4. Reconnect: urgent notes go first
5. Two devices disagree, a person decides

ALSO IN THE BOX

Hands-free dictation in any language, on the device · Private text sealed on disk · First PIN claimed with a console code · Devices learn from each other when the cloud is gone · Photos searchable by what they show · Encrypted backup to a replacement device · Shift report · One-click staged conflict · Installable on a phone · Hindi and other languages optional · /healthz and /metrics

NEXT

TLS for remote dashboards · Faces and name plates in photos · Photos between peers

Try it live

LIVE DEVICE

fieldmindedgea.hidenplay.net

edge-a, running 24/7 on an arm64 server with Qdrant Cloud. Ask us for the PIN, then click **Demo guide**. A second device, edge-b, syncs with it.

CODE AND TESTS

github.com/yabhinav1/fieldmind

MIT. 132 tests on every push against a real Qdrant Server and a real browser. `docker compose --profile demo up` runs the whole thing on any laptop.

FIVE THINGS TO TRY IN TWO MINUTES

Turn the Network switch off and keep working · Click Dictate and just talk, in any language · Ask “is a vibration of 7.2 mm/s acceptable” · Type a note with a name and a phone number and watch the preview mask them · Demo guide → Stage a conflict, then decide